

Exploratory Data Analysis

MovieLens 100k - Collaborative Filtering Dataset

Source: GroupLens Research, files.grouplens.org/datasets/movielens/ml-100k.zip

Generated: 2026-09-19

Dataset Overview & Cleaning Summary

ratings: (100000, 4) | movies: (1682, 25) | users: (943, 5)

```
ratings.dtypes:
user_id          int64
item_id          int64
rating           float64
timestamp        datetime64[s]
```

```
ratings.describe():
count    100000.00
mean       3.53
std        1.13
min         1.00
25%         3.00
50%         4.00
75%         4.00
max         5.00
```

duplicates removed / orphan keys found: 0 (verified above)

```
users.describe() (age):
count     943.0
mean      34.1
std       12.2
min        7.0
25%       25.0
50%       31.0
75%       43.0
max       73.0
```

```
gender counts:
gender
M     670
F     273
```

```
top 10 occupations:
occupation
student          196
other            105
educator          95
administrator     79
engineer          67
programmer        66
librarian         51
writer            45
executive         32
scientist         31
```

Ratings Analysis

rating value counts:

```
rating
1.0    6110
2.0   11370
3.0   27145
4.0   34174
5.0   21201
```

ratings per user - describe():

```
count    943.0
mean     106.0
std       100.9
min        20.0
25%        33.0
50%        65.0
75%       148.0
max       737.0
```

ratings per movie - describe():

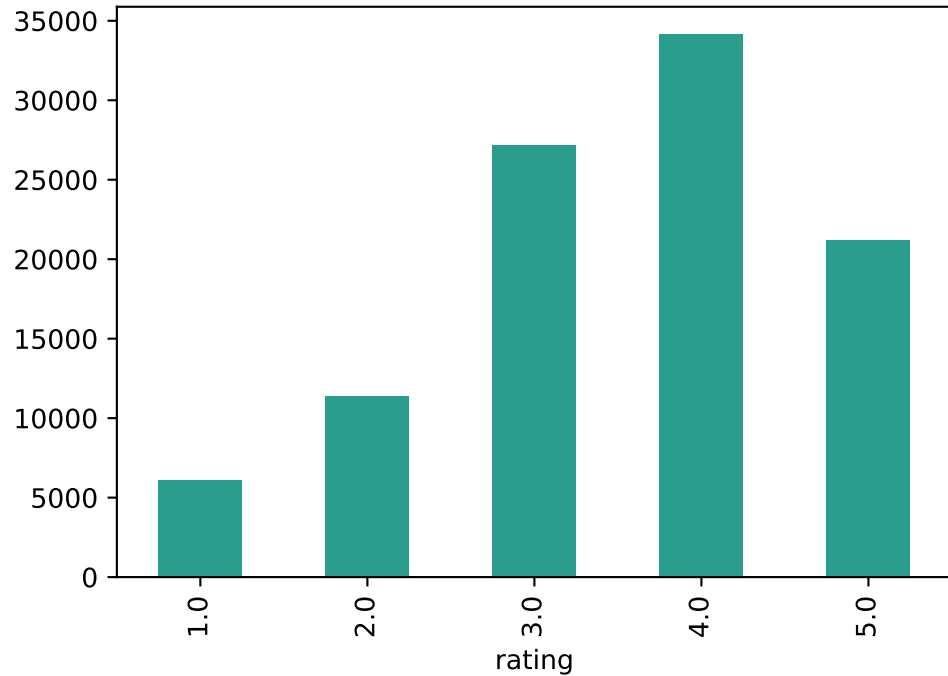
```
count   1682.0
mean     59.5
std       80.4
min        1.0
25%        6.0
50%       27.0
75%       80.0
max      583.0
```

matrix sparsity: 0.937 (fraction of the 943x1682 user-item matrix that is empty)

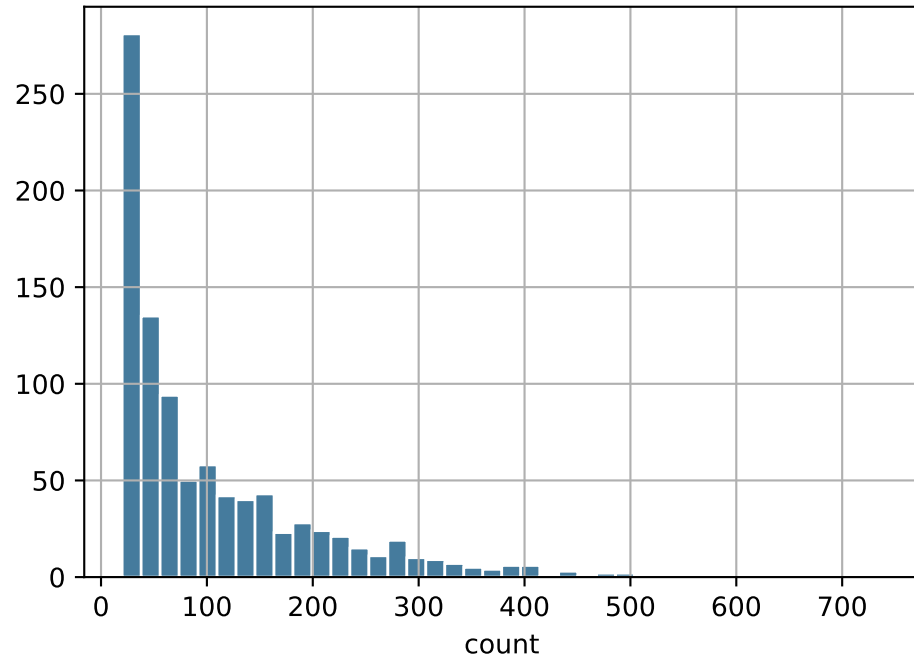
top 10 most-rated movies:

	n_ratings	title
item_id		
50	583	Star Wars (1977)
258	509	Contact (1997)
100	508	Fargo (1996)
181	507	Return of the Jedi (1983)
294	485	Liar Liar (1997)
286	481	English Patient, The (1996)
288	478	Scream (1996)
1	452	Toy Story (1995)
300	431	Air Force One (1997)
121	429	Independence Day (ID4) (1996)

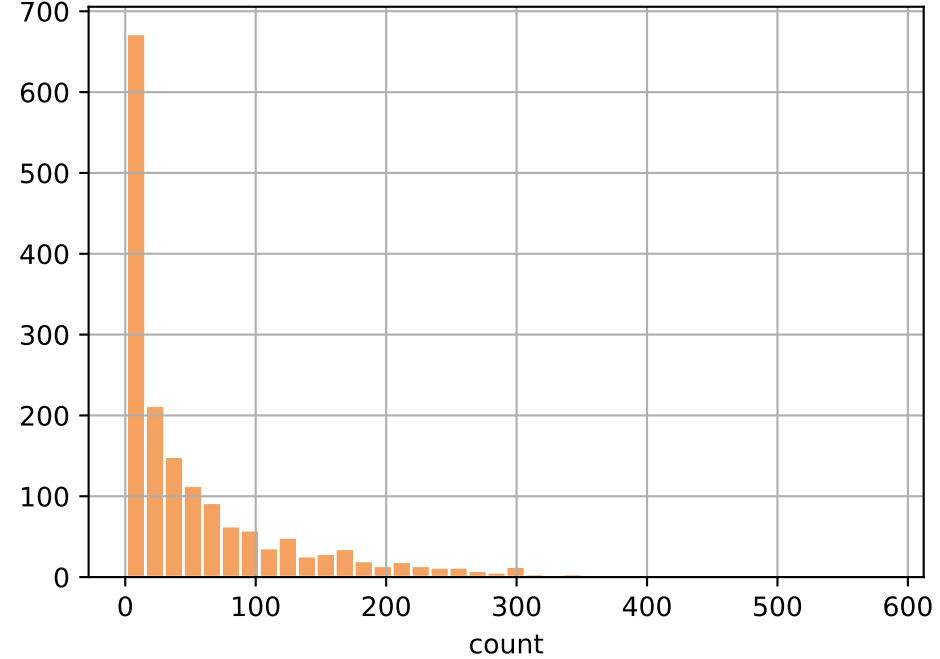
Rating distribution



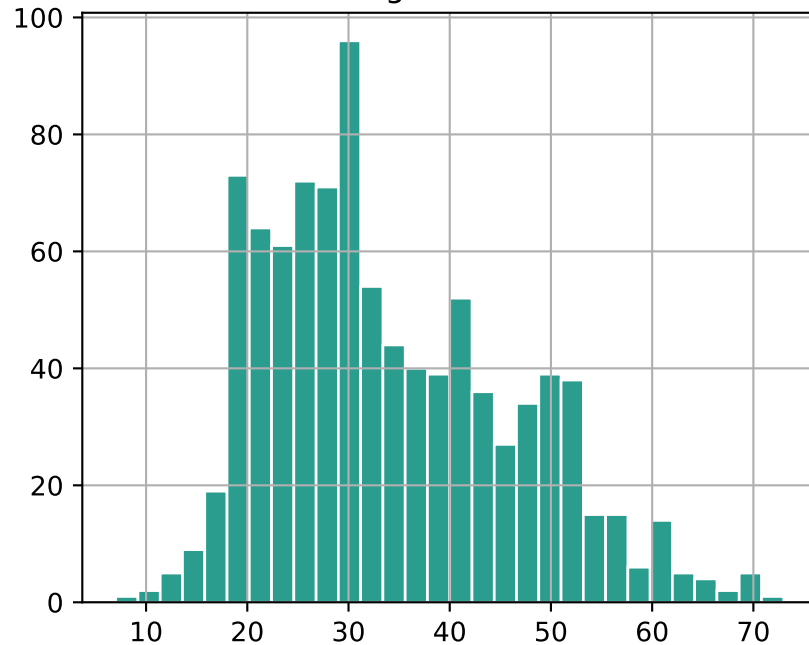
Ratings per user



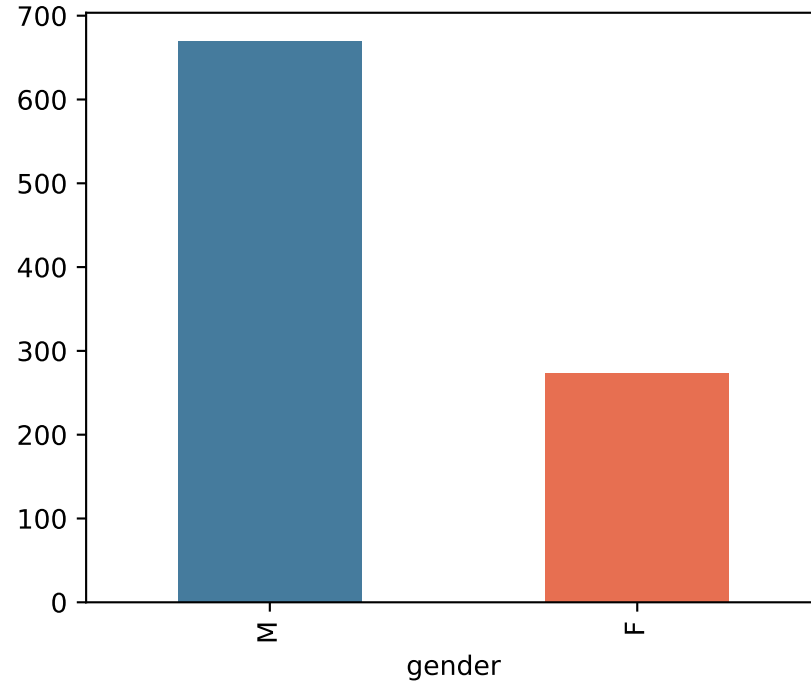
Ratings per movie



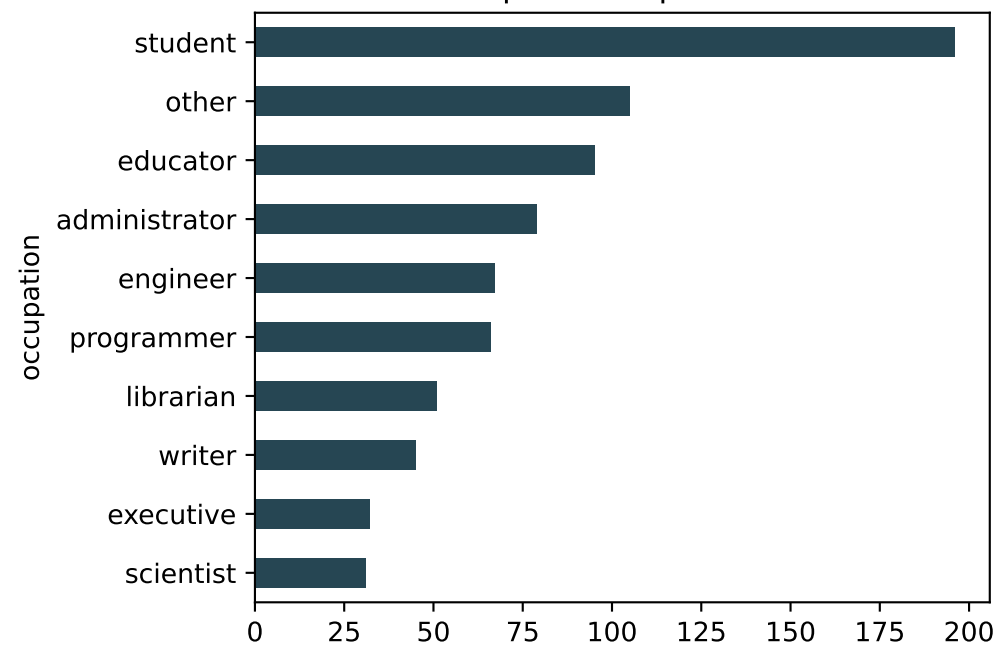
User age distribution



Gender counts



Top 10 occupations



Genre Analysis

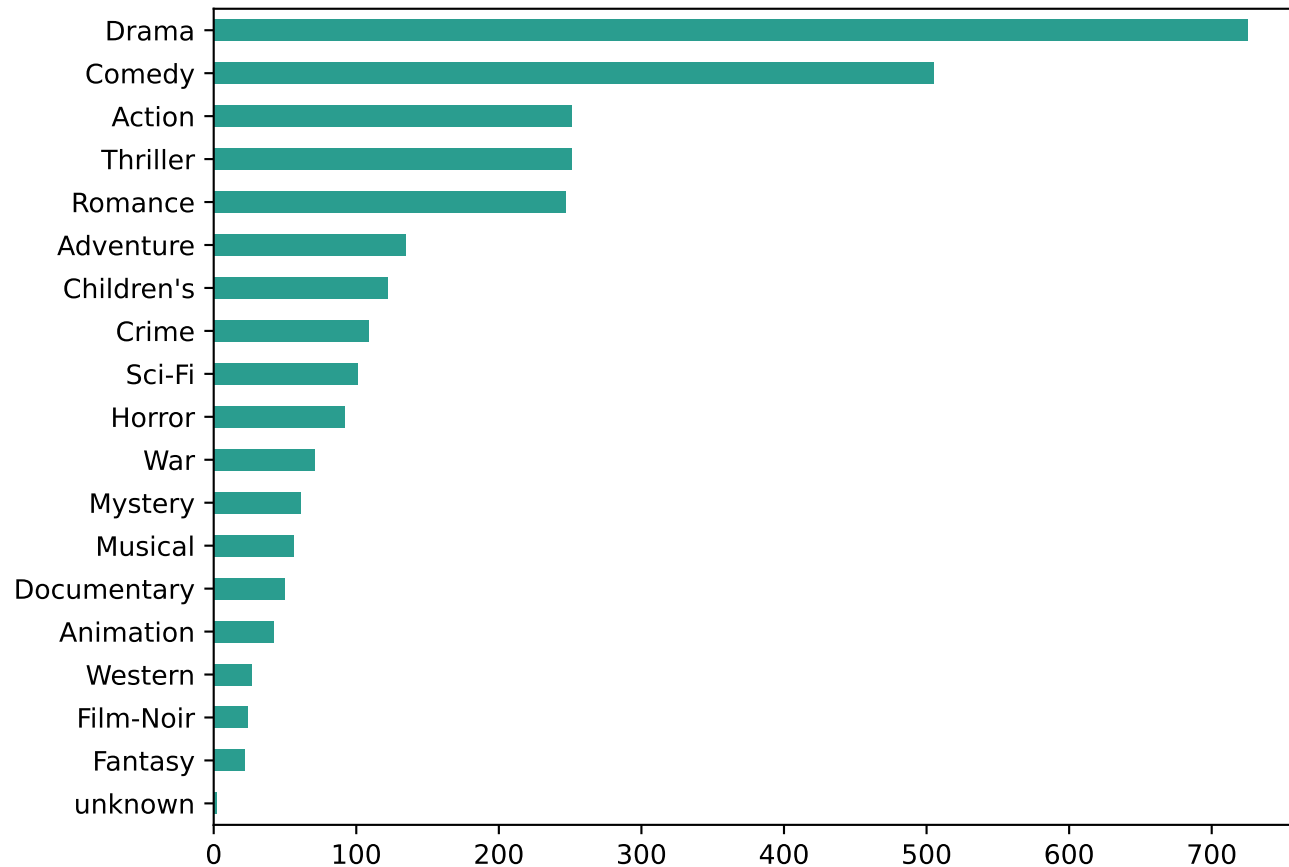
movie count by genre:

Drama	725
Comedy	505
Action	251
Thriller	251
Romance	247
Adventure	135
Children's	122
Crime	109
Sci-Fi	101
Horror	92
War	71
Mystery	61
Musical	56
Documentary	50
Animation	42
Western	27
Film-Noir	24
Fantasy	22
unknown	2

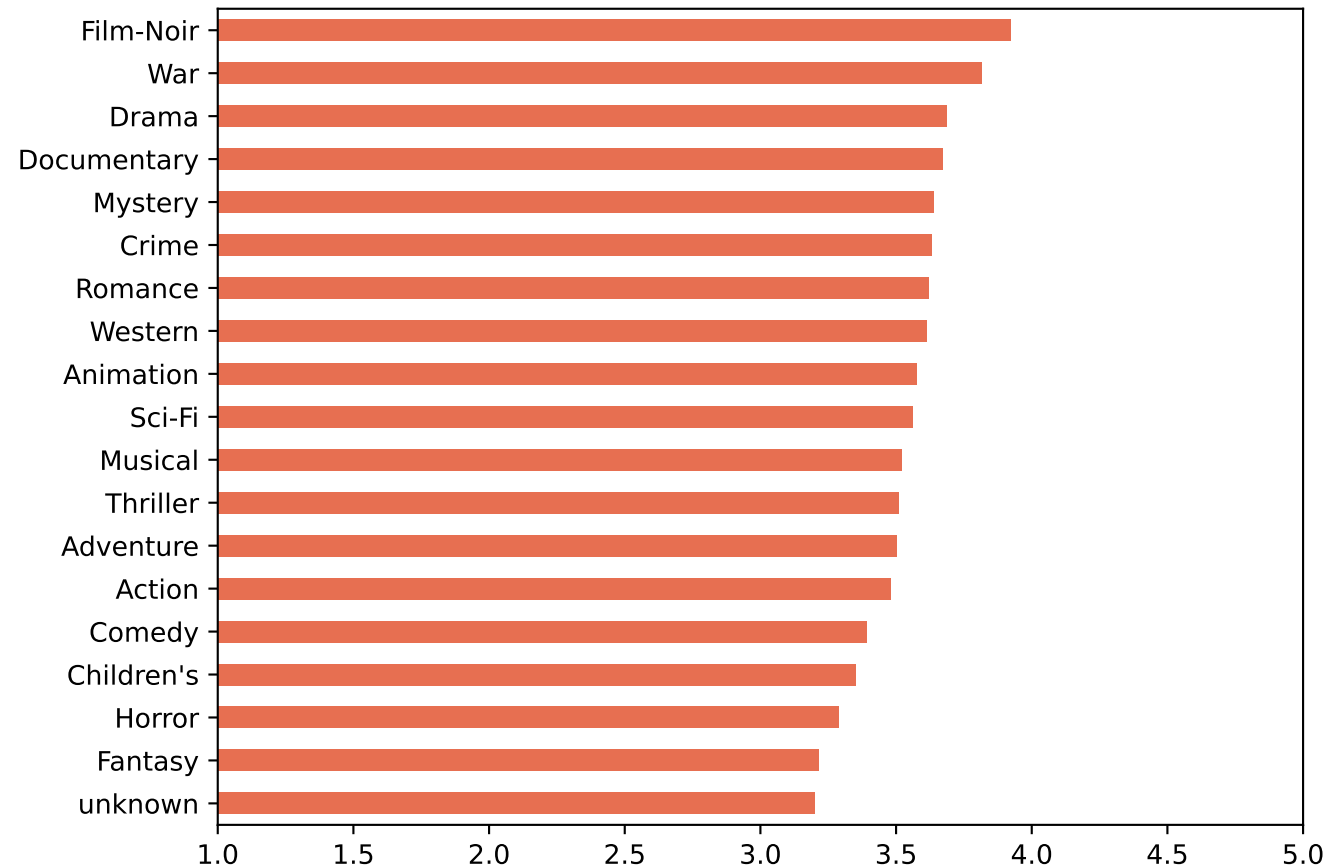
average rating by genre:

Film-Noir	3.922
War	3.816
Drama	3.687
Documentary	3.673
Mystery	3.638
Crime	3.632
Romance	3.622
Western	3.613
Animation	3.577
Sci-Fi	3.561
Musical	3.521
Thriller	3.509
Adventure	3.504
Action	3.480
Comedy	3.394
Children's	3.353
Horror	3.290
Fantasy	3.215
unknown	3.200

Number of movies per genre



Average rating per genre



Ratings submitted per week

